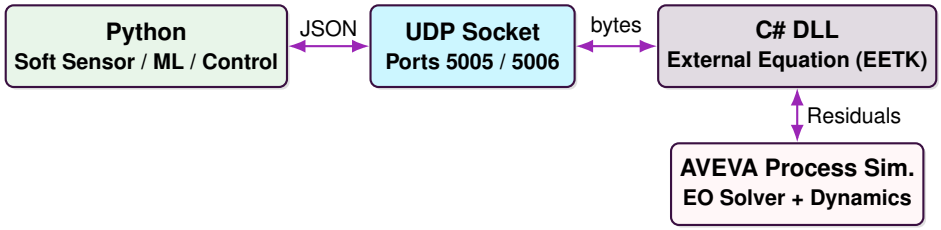
OPEN-SOURCE ENGINEERING GUIDE

A complete guide to building real-time co-simulation bridges between AVEVA's Equation-Oriented solver and external Python controllers, enabling soft sensors, advanced control, and digital twin telemetry.



Author

Ezequiel José Valencia Urbina

Mechatronics Engineer
M.Sc. Candidate – Mechanical Systems Dynamics
University of Brasília (UnB), Brazil
urbina.ezequiel@aluno.unb.br

Video Tutorial: https://youtu.be/r88qSzDYIMg
GitHub Repository: https://github.com/VzequyU/AVEVA-Process-Simulation-UDP
Date: August 2026 – Version 2.0

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1 Introduction and System Architecture

Building an interactive digital twin requires connecting rigorous thermodynamic and hydraulic models with external analysis environments (such as Python) in real time. This document provides a complete, step-by-step guide to achieve bidirectional co-simulation using **AVEVA Process Simulation**, the **External Equation Toolkit (EETK)** through a C# DLL, and **UDP socket** communication.

1.1 Who Is This Guide For?

- **Process engineers** who want to connect AVEVA simulations to Python-based analytics.
- **Control engineers** building advanced control loops or soft sensors.
- **Researchers** developing digital twins for refineries, petrochemical plants, or green hydrogen systems.
- **Graduate students** exploring equation-oriented simulation and real-time data integration.

1.2 What You Will Build

By the end of this guide, you will have a working system where:

1. A Python script sends commands (e.g., valve opening setpoints) to AVEVA in real time.
2. AVEVA's dynamic solver processes these commands through your custom DLL equations.
3. The DLL sends telemetry data (flows, levels, temperatures) back to Python for visualization or ML inference.

1.3 The Degrees of Freedom Principle

In an equation-oriented (EO) simulator, you cannot simply force a variable to a value. You must interact *mathematically* through the **Degrees of Freedom (DOF)**.

If we want to control the opening of a valve (`ManPos`) from Python, we must *unspec* that variable in AVEVA (unchecking its specification box) to create an additional degree of freedom. Then, our DLL injects a new equation (residual) that closes the mathematical balance:

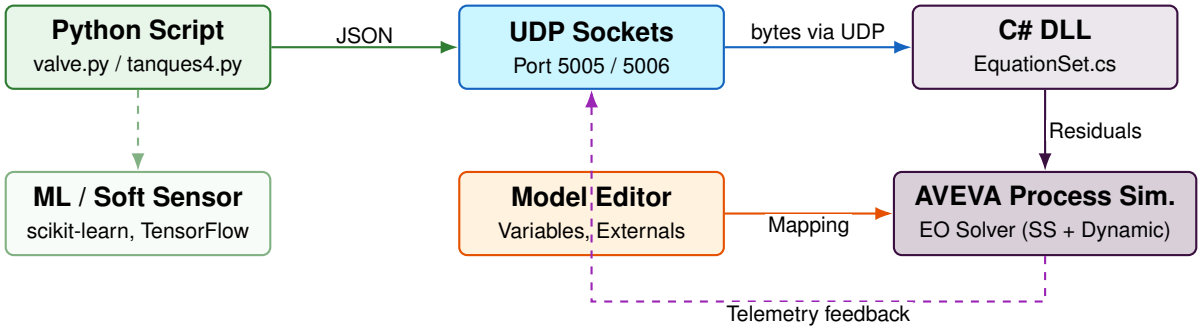
$$\text{Residual} = \text{ManPos}_{\text{AVEVA}} - \text{Target}_{\text{Python}} = 0 \tag{1}$$

When the solver converges, the valve position matches the Python setpoint exactly.

⊕ TECHNICAL NOTE

Why UDP instead of TCP? UDP is connectionless and non-blocking by design, which is critical inside a numerical solver that may call your equation function thousands of times per second. A blocking TCP call would freeze the entire simulation. UDP packets that arrive late are simply discarded, which is acceptable for telemetry.

1.4 Architecture Overview



1.5 Prerequisites

Software	Version	Purpose
AVEVA Process Simulation	2023 or later	Process simulator with EO solver
Visual Studio	2019 / 2022	C# DLL compilation
EETK (External Equation Toolkit)	Included with AVEVA	SDK, headers, templates
Python	3.8+	External controller / analytics

2 Phase 1: Visual Studio Project Setup

The first step is building the mathematical bridge (the DLL) using the official templates provided by AVEVA.

2.1 Locating the EETK

The External Equation Toolkit is installed alongside AVEVA Process Simulation. Look for it in:

`C:\Program Files\AVEVA\Process Simulation\EETK\`

or within your user profile if installed per-user. The EETK folder contains:

Item	Description
`SimpleEquationSet/`	Template project for scalar equations (start here)
`ArrayEquationSet/`	Template for vector/multicomponent equations
`CustomReactionRateSample/`	Example for custom kinetics
`SimSci.SimCentral.Interfaces.dll`	The interface library your DLL must reference
`ExternalEquation.lib`	Import library for the solver bridge
`IUserPackage.hpp`	Header file (C++ interface, not needed for C#)

2.2 Creating the C# Project

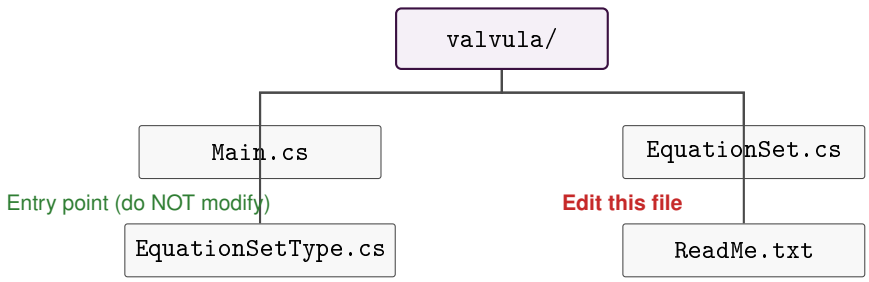
1. Open **Visual Studio 2019+** and select *Create a new project*.
2. Search for the `SimpleEquationSet` template provided by the EETK. Alternatively, copy the entire `SimpleEquationSet` folder and rename it (e.g., `valvula`).
3. Assign a name to the project (e.g., `valvula` for a valve controller).

△ CRITICAL WARNING

Critical Path Requirement: Save the project **strictly inside the EETK folder** (e.g., `%userprofile%\source\repos\EETK\`). Check *"Place solution and project in the same directory"*. If you place it elsewhere, the relative references to `SimSci.SimCentral.Interfaces` will break and the project will not compile.

2.3 Project Structure

Your project should contain these key files:



⊕ TECHNICAL NOTE

Main.cs implements `IExternalClass` and is the entry point AVEVA calls when loading the DLL. It creates instances of `EquationSet`. **Never modify Main.cs** — all your custom logic goes in `EquationSet.cs`.

2.4 Verifying References

In the Solution Explorer, expand **References**. You must see:

- `SimCentral.Server.ExternalEquation.Interfaces` — this provides `IExternalEquationSet`, `Severity`, `DataType`, etc.
- Standard .NET references: `System`, `System.Core`, `System.Net.Http`.

If the reference has a yellow warning icon (△), right-click → Remove, then Add Reference → Browse and locate the DLL in the EETK folder.

3 Phase 2: The C# DLL — Equations and UDP Telemetry

The core of our connector resides in `EquationSet.cs`. This code will instantiate a UDP client, listen for commands from Python on port 5006, modify the solver residuals, and send telemetry responses on port 5005.

3.1 Example 1: Single Valve Controller

This example controls the opening of a single valve (`XV1`) from Python and reads the mass flow at a sink (`SNK1`).

```
using SimSci.SimCentral.Interfaces;
using System;
using System.Collections.Generic;
using System.Net.Sockets;
using System.Text;
using System.Net;

namespace SimCentral.Examples.valvula
{
    public sealed class EquationSet : IExternalEquationSet
    {
        // --- EQUATION AND VARIABLE DECLARATION ---
        private static IReadOnlyList<string> Equations = new[] { "ControlManPos" };
        private static IReadOnlyList<string> EqnText = new[] { "Valve opening control from
    Python" };
        private static IReadOnlyList<string> Variables = new[] { "ManPos", "FlujoSink" };
        private const int _MANPOS = 0, _FLUJOSINK = 1;

        // INCIDENCE MATRIX: declare ALL variables to avoid singularity
        private static readonly int[][] Incidence = { new[] { _MANPOS, _FLUJOSINK } };

        private readonly double[][] derivatives;
        private string lastMessage;
        private Severity lastSeverity;
        private readonly double[] v;

        // --- UDP NETWORKING ---
        private static UdpClient udpClient;
        private static IPEndPoint pythonEndPoint;
        private double targetPos = 0.5; // Default 50% open

        public EquationSet(string nameIn)
        {
            this.v = new double[Variables.Count];
            this.derivatives = new double[Equations.Count][];
            this.derivatives[0] = new double[Incidence[0].Length];
            this.ClearErrors();

            if (udpClient == null)
            {
                try {
                    udpClient = new UdpClient(5006);
                    pythonEndPoint = new IPEndPoint(
                        IPAddress.Parse("127.0.0.1"), 5005);
                } catch { }
            }
        }

        public Severity GetEquationResidual(int equationNumber,
            out double residual, bool calculateDerivative)
        {
```

```
            // 1. LISTEN TO PYTHON (non-blocking)
            if (udpClient != null && udpClient.Available > 0)
            {
                try {
                    IPEndPoint remoteEP = new IPEndPoint(IPAddress.Any, 0);
                    byte[] rcvBytes = udpClient.Receive(ref remoteEP);
                    string rcvMsg = Encoding.UTF8.GetString(rcvBytes);
                    if (double.TryParse(rcvMsg,
                        System.Globalization.NumberStyles.Any,
                        System.Globalization.CultureInfo.InvariantCulture,
                        out double newVal))
                    {
                        targetPos = Math.Max(0.0, Math.Min(1.0, newVal));
                    }
                } catch { }
            }

            // 2. RESIDUAL FOR THE SOLVER
            residual = v[_MANPOS] - targetPos;

            if (calculateDerivative)
            {
                this.derivatives[0][0] = 1.0;   // dRes/dManPos
                this.derivatives[0][1] = 0.0;   // dRes/dFlujoSink
            }

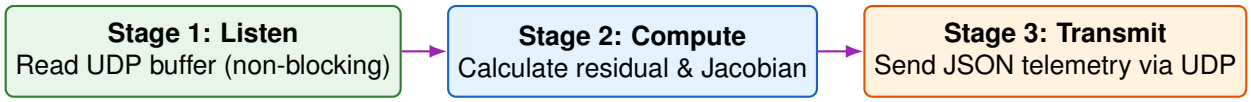
            // 3. SEND TELEMETRY TO PYTHON
            if (!calculateDerivative && udpClient != null)
            {
                try {
                    string json = string.Format(
                      "{{\"flujo_sink\": {0:F4}, \"manpos_actual\": {1:F4}}}",
                      v[_FLUJOSINK], v[_MANPOS]);
                    byte[] bytes = Encoding.UTF8.GetBytes(json);
                    udpClient.Send(bytes, bytes.Length, pythonEndPoint);
                } catch { }
            }
            return Severity.Ok;
        }

        // ... (standard interface methods omitted for brevity)
    }
}
```

Listing 1: Complete EquationSet.cs for bidirectional valve control

3.1.1 Understanding the Three Stages

Every call to `GetEquationResidual` follows a three-stage pattern:



★ ENGINEERING TIP

Why send telemetry only when `calculateDerivative == false`**?** During a single solver iteration, AVEVA calls `GetEquationResidual` multiple times — once for the residual value and several times for numerical Jacobian perturbations. Sending UDP packets on every call would flood the network. By filtering on `!calculateDerivative`, we send data only on the

"real" evaluation.

3.2 Example 2: Quadruple-Tank System (Johansson)

The second example demonstrates a more complex scenario: controlling two inlet flows (Q_1, Q_2) and reading four tank levels (H_1 through H_4) in a classic Johansson quadruple-tank benchmark.

```
namespace SimCentral.Examples.tanques4
{
    public sealed class EquationSet : IExternalEquationSet
    {
        // 2 Equations to inject flows Q1 and Q2
        private static IReadOnlyList<string> Equations =
            new[] { "ControlQ1", "ControlQ2" };
        private static IReadOnlyList<string> Variables =
            new[] { "Q1", "Q2", "H1", "H2", "H3", "H4" };
        private const int _Q1=0, _Q2=1, _H1=2, _H2=3, _H3=4, _H4=5;

        // Include ALL variables in both equations
        private static readonly int[][] Incidence = {
            new[] { _Q1, _Q2, _H1, _H2, _H3, _H4 },
            new[] { _Q1, _Q2, _H1, _H2, _H3, _H4 }
        };

        // ... (UDP setup identical to Example 1)

        public Severity GetEquationResidual(int equationNumber,
            out double residual, bool calculateDerivative)
        {
            // Listen on first equation only (avoid double-read)
            if (equationNumber == 0 && udpClient != null
                && udpClient.Available > 0)
            {
                // Parse JSON: {"q1": 0.08, "q2": 0.05}
                // ... (parsing code)
            }

            // Residual per equation
            if (equationNumber == 0)
                residual = v[_Q1] - targetQ1;
            else
                residual = v[_Q2] - targetQ2;

            // Send all 4 tank levels to Python
            if (equationNumber == 1 && !calculateDerivative)
            {
                // JSON: {"h1":..., "h2":..., "h3":..., "h4":...}
            }
            return Severity.Ok;
        }
    }
}
```

Listing 2: EquationSet.cs for the Johansson 4-tank system

◇ INDUSTRY BEST PRACTICE

When you have **multiple equations**, listen to Python only on `equationNumber == 0` and send telemetry only on the last equation. This prevents reading the same UDP packet twice and avoids sending duplicate data per solver iteration.

4 Phase 3: AVEVA Process Simulation Configuration

4.1 Deploying the DLL

After a successful build in Visual Studio (`Build` → `Rebuild Solution`), copy the resulting `.dll` file to:

`...\SimCentral Addons\Engineering\`

△ CRITICAL WARNING

The Phantom DLL Problem: Windows locks DLL files in memory. If you replace the `.dll` while AVEVA is open, the simulator will continue using the old version from its RAM cache. **Always close AVEVA completely** before replacing the DLL file.

4.2 The Intermediary Block Approach

The recommended architecture encapsulates the DLL in a custom Model Type (`TestModel`) that acts as a controller block alongside your process equipment.

STEP 1 – Create the Model Type

Open the **Model Editor** (from the Library panel). Create a new Model Type called `TestModel`. This block will host your external equations.

STEP 2 – Declare Variables

In the **Variables** section, add the variables that match your C# code *exactly by name*:

Name	Type	Units	Default	Min / Max
`ManPos`	Fraction	fraction	0.5	0 / 1
`FlujoSink`	Mass Flow	kg/s	1	-10^6 / 10^6

STEP 3 – Configure External DLL

Expand the **Externals** section and click **Add External** (+).

Field	Value
Condition	UseExt
Name	Ext1
Dll	`valvulas.dll`
Implementation Class	`SimCentral.Examples.valvula.Main`

The **Implementation Class** follows the pattern: `<Namespace>.Main`, where the namespace matches your C# code exactly.

STEP 4 – Set Equations to External

In the Model Type properties, change the **Equations** dropdown from *Internal* to **External**. This tells AVEVA to use your DLL instead of built-in equations.

STEP 5 – Build the Flowsheet

Drag an instance of your TestModel (`M1`) onto the simulation canvas alongside your process equipment (e.g., `SRC1` → `XV1` → `SNK1`).

STEP 6 – Release Degrees of Freedom

This is the most critical step. Go to the valve `XV1` and **uncheck** (unspec) `ManPos`. Also uncheck `M1.ManPos` and `M1.FlujoSink`. This creates the degrees of freedom that your DLL equations will fill.

STEP 7 – Create Bridge Equations

On the simulation canvas, add equations that link the Model Type variables to the process equipment:

```
Eqn1:    XV1.ManPos = M1.ManPos
Eqn2:    M1.FlujoSink = SNK1.W
```

These equations connect the DLL's calculated `ManPos` to the physical valve, and feed the sink's mass flow back to the DLL for telemetry.

STEP 8 – Solve

Click **Solve**. If everything is configured correctly, all variables will turn green (Status = Solved).

⊘ TROUBLESHOOTING

Singularity Error (Red Crosses): If the canvas fills with red crosses after switching to *External*, AVEVA detected variables declared in C# that do not participate in the Incidence Matrix. **Solution:** Include *all* mapped variables in the `Incidence` array, even if their derivative is 0.0. For example, `FlujoSink` must appear in the incidence even though it is only read, not computed.

5 Phase 4: The Python Controller

With the simulation ready, switch AVEVA to **Dynamics** mode and press **Play**. Then run the Python script to close the control loop.

5.1 Valve Controller Script

```python
import socket
import json
import threading
import time

# Network configuration (reversed from C#)
UDP_IP = "127.0.0.1"
PORT_RECEIVE = 5005   # Python listens here (AVEVA sends here)
PORT_SEND = 5006      # Python sends here (AVEVA listens here)

sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.bind((UDP_IP, PORT_RECEIVE))
sock.setblocking(False)

# Background thread: listen for AVEVA telemetry
def listen_aveva():
    while True:
        try:
            data, addr = sock.recvfrom(1024)
            msg = json.loads(data.decode('utf-8'))
            print(f"[AVEVA] Flow: {msg['flujo_sink']:.4f} kg/s "
                  f"| Valve: {msg['manpos_actual']:.2f}")
        except BlockingIOError:
            time.sleep(0.05)
        except Exception:
            pass

threading.Thread(target=listen_aveva, daemon=True).start()

# Main loop: send commands
print("=== AVEVA <-> Python Control Panel ===")
print("Enter a value between 0.0 and 1.0 (or 'q' to quit)")

while True:
    cmd = input("Valve position: ")
    if cmd.lower() == 'q':
        break
    try:
        val = float(cmd)
        sock.sendto(str(val).encode('utf-8'), (UDP_IP, PORT_SEND))
        print(f">>> Command {val} sent to AVEVA.")
    except ValueError:
        print("Please enter a valid number.")
```

Listing 3: valve.py — Interactive valve controller via UDP

5.2 Real-Time Visualization (4-Tank Example)

For the quadruple-tank system, the Python script includes `matplotlib` animation for live level monitoring:

```python
import matplotlib.pyplot as plt
import matplotlib.animation as animation
```

```
from collections import deque

MAX_PTS = 200
h1_data, h2_data = deque(maxlen=MAX_PTS), deque(maxlen=MAX_PTS)
h3_data, h4_data = deque(maxlen=MAX_PTS), deque(maxlen=MAX_PTS)

fig, axs = plt.subplots(2, 2, figsize=(10, 6))
fig.suptitle('AVEVA Digital Twin - Dynamic Response')

def update(frame):
    # Plot each tank level in its subplot
    for ax, data, color, name in zip(
        axs.flat, [h1_data, h2_data, h3_data, h4_data],
        ['b','r','g','m'], ['Tank 1','Tank 2','Tank 3','Tank 4']):
        ax.clear()
        ax.plot(data, f'{color}-', lw=2)
        ax.set_title(name)
        ax.grid(True, alpha=0.3)

ani = animation.FuncAnimation(fig, update, interval=100)
plt.show()
```

Listing 4: tanques4.py — Real-time 4-tank level visualization (excerpt)

◇ INDUSTRY BEST PRACTICE

Thread Safety: Using `socket.setblocking(False)` with `threading` ensures the Python command console does not freeze while listening to the 20ms telemetry stream from AVEVA's dynamic solver iterations.

6 Quick-Start Checklist

#	Action	Done
1	Locate EETK folder in AVEVA installation	☐
2	Copy `SimpleEquationSet` template; rename to your project	☐
3	Open in Visual Studio; verify `Interfaces` reference is green	☐
4	Edit `EquationSet.cs` with your equations + UDP code	☐
5	Build → Rebuild Solution (x64, Release)	☐
6	Close AVEVA completely	☐
7	Copy `.dll` to `SimCentral Addons\Engineering\`	☐
8	Create Model Type with matching variable names	☐
9	Add External reference: DLL name + Implementation Class	☐
10	Set Equations to External	☐
11	Unspec controlled variables; add bridge equations	☐
12	Solve (steady-state) → all green	☐
13	Run `python valve.py`	☐
14	Switch AVEVA to Dynamics → Play	☐
15	Send commands from Python; verify response	☐

7 Common Errors and Solutions

⊘ TROUBLESHOOTING

CS0246: Type "Severity" could not be found
The project lost its reference to `SimCentral.Server.ExternalEquation.Interfaces`. In Solution Explorer, check if the reference has a yellow warning icon. Remove it and re-add via Browse, pointing to the DLL in the EETK folder.

⊘ TROUBLESHOOTING

AVEVA shows "DLL not found" or External status is red
Verify: (1) the DLL is in the correct `Engineering` subfolder of `SimCentral Addons`; (2) the **Implementation Class** matches the exact namespace + `.Main` (e.g., `SimCentral.Examples.valvula.Main`); (3) you restarted AVEVA after copying the DLL.

⊘ TROUBLESHOOTING

Singularity / Red crosses after switching to External
All variables declared in `VariableNames` must appear in the `Incidence` matrix, even if their derivative is zero. Add missing variables to each equation's incidence array.

⊘ TROUBLESHOOTING

Python receives no data from AVEVA
Check: (1) Python binds to port 5005 *before* starting the simulation; (2) the C# DLL sends to port 5005; (3) no firewall is blocking localhost UDP; (4) AVEVA is in **Dynamics mode** and running (Play).

⊘ TROUBLESHOOTING

Valve does not respond to Python commands
Verify that `XV1.ManPos` is **unspecified** (unchecked). If it remains specified, AVEVA's own specification overrides the DLL equation, making it a redundant/conflicting constraint.

8 Extending the System: Soft Sensors and ML

The architecture presented here is the foundation for more advanced applications:

- **Soft Sensors:** Use the telemetry stream to feed a machine learning model (e.g., scikit-learn, TensorFlow) that predicts unmeasured variables like product quality or composition.
- **Model Predictive Control (MPC):** Replace the simple setpoint injection with an MPC controller that optimizes valve positions based on predicted future states.
- **AVEVA PI System Integration:** Write the telemetry data to PI tags for historian storage and enterprise-wide visibility.
- **Green Hydrogen Systems:** Apply the same DLL/UDP architecture to electrolyzer and fuel cell models for real-time optimization of hydrogen production.
- **Refinery Inference:** Deploy trained regression models that infer distillation column temperatures or product octane numbers from available sensor data.

◇ INDUSTRY BEST PRACTICE

The code examples and pre-configured simulation files for both the valve and quadruple-tank systems are available in the GitHub repository. Download, extract, and start experimenting.

9 Resources and Contact

Video Tutorial	https://youtu.be/r88qSzDYIMg
GitHub Repository	https://github.com/AVEVA-Process-Simulation-UDP
Author Email	urbina.ezequiel@aluno.unb.br
University	University of Brasília (UnB) – Mechanical Sciences

If this guide helped you, consider starring the GitHub repository and sharing it with your colleagues. Contributions and pull requests are welcome.